SRI KRISHNA INSTITUTE OF TECHNOLOGY

(Accredited by NAAC, Approved by A.I.C.T.E. New Delhi, Recognised by Govt. of Karnataka & Affiliated to V.T U., Belagavi)
#29, Chimney Hills, Hesaraghatta Main Road, Chikkabanavara Post, Bengaluru- 560090

Department of Artificial Intelligence and Machine Learning

Subject Name: Data Science & Its Applications
Subject Code: 21AD62

SEM: 6th **DIV: A**

Faculty: Prof. Manzoor Ahmed

Module-2 Question Bank

SL#	Question	CO	Level	Marks
1.	Explain null and alternative hypothesis by considering the example for a flipping coin.	CO2	L3	8
2.	Describe A/B test with an example.	CO2	L3	8
3.	Explain how gradient descent is used to fit parameterized models.	CO2	L3	8
4.	Explain p-Values with an example.	CO2	L2	8
5.	Describe Bayesian inference in detail.	CO2	L3	8
6.	Explain Confidence Intervals and p-hacking with an example.	CO2	L3	8
7.	Explain how gradient is estimated in detail.	CO2	L3	8
8.	Explain how gradient descent is use to fit the models.	CO2	L3	8
9.	What is the drawback of gradient descent and explain how to overcome the drawback with an example.	CO2	L3	8
10.	Explain cleaning and munging of data with an example.	CO2	L3	8
11.	Illustrate web scraping using Python programming by considering the below hypothetical scenario. The VP of Policy at DataSciencester is worried about potential regulation of the data science industry and asks you to quantify what Congress is saying on the topic. In particular, he wants you to find all the representatives who have press releases about “data”.	CO2	L3	10
12.	Write a python code for scrapping an HTML document with an example	CO2	L2	6
13.	Explain the scraping of data using APIs with example.	CO2	L3	8
14.	Describe how data can be manipulated by considering an example.	CO2	L3	8
15.	Write Python program to extract the date and time of a GitHub repository created and also the language of the last five repositories.	CO2	L3	8
16.	Explain the reading of basic text files with example.	CO2	L2	8
17.	Illustrate tqdm library by considering an example.	CO2	L3	8
18.	Illustrate the difference between named tuples and Data classes with example.	CO2	L2	8
19.	Explain exploring of one dimensional data and plot normal and uniform distribution histogram with an example.	CO2	L3	10
20.	Explain exploring of two dimensional data and find out the correlation between the two dimensions with an example.	CO2	L3	10
21.	Explain in detail dimensionality reduction with example.	CO2	L3	10

Faculty Signature